

Homework 2: DAGs and E-Values - Solution

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1 Literature Review

Perform a literature review on variables/factors confounding the association between IVH and mortality.

Based on the literature review, several key risk factors were identified that are associated with both Intraventricular Hemorrhage (IVH) and Neonatal Mortality in Very Low Birth Weight (VLBW) infants.

Publications (Example Sources)	Potential Confound- ing Factors	Association w/ IVH	Estimates (95% CI)	Association w/ Mortality	Estimates (95% CI)
<i>Wei et al. (2016) / Roberts et al. (2017)</i>	Antenatal Corticosteroids (ACS)	Protective (Reduces IVH risk)	Severe IVH: OR = 0.51 (0.45-0.58) [1]	Protective (Reduces Mortality)	Neonatal death: RR = 0.69 (0.58-0.81) [2]
<i>Villamor-Martinez et al. (2018)</i>	Chorioamnion (Clinical/Histological)	Risk Factor (Increases IVH risk)	OR = 1.88 (1.61-2.19) [3]	Associated with prematurity/sepsis pathways	(Estimates vary by outcome definition/study)
<i>HW1 Data / Literature</i>	Gestational Age	Strong Negative Assoc.	(See HW1 results)	Strong Negative Assoc.	(See HW1 results)
<i>HW1 Data / Literature</i>	Birth Weight	Strong Negative Assoc.	(See HW1 results)	Strong Negative Assoc.	(See HW1 results)
<i>Zhao et al. (2022) / Lee et al. (2010)</i>	Low 5-min Apgar score	Risk Factor (Increases IVH risk)	Apgar ≤ 7 : OR = 2.273 (1.163-4.442) [4]	Risk Factor (Increases Mortality)	Example (24 wks): RR = 3.1 (2.9-3.4) for Apgar 0-3 vs 7-10 [5]

2 Directed Acyclic Graph (DAG)

Draw a Directed Acyclic Graph (DAG) to illustrate the conceptual framework of the causal association between IVH and infant death.

Below is the DAG written in **DAGitty syntax** (copy/paste into <https://dagitty.net/dags.html>):

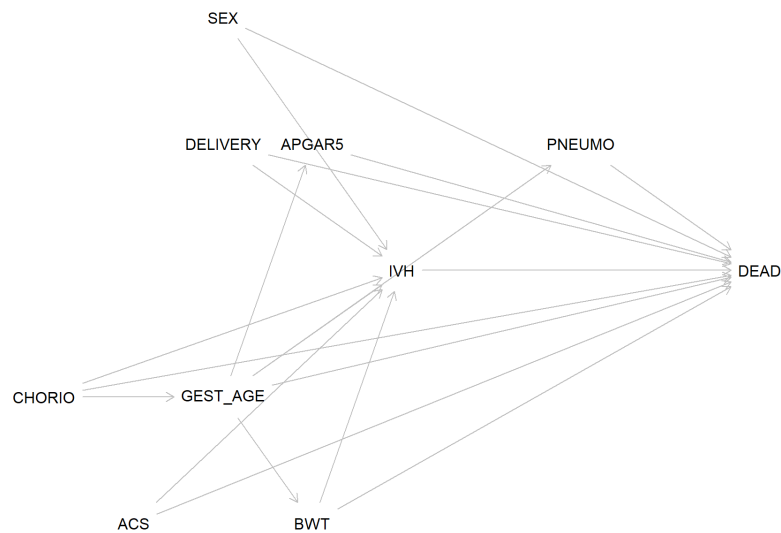


Figure 1: Final DAG for IVH and mortality

DAGitty Code:

```

dag {
  bb="-1,-1,9,6"

  IVH [exposure,pos="4,3"]
  DEAD [outcome,pos="8,3"]
  GEST_AGE [pos="2,4"]
  BWT [pos="3,5"]
  ACS [pos="1,5"]
  CHORIO [pos="0,4"]
  DELIVERY [pos="2,2"]
  APGAR5 [pos="3,2"]
  SEX [pos="2,1"]
  PNEUMO [pos="6,2"]

  IVH -> DEAD
  IVH -> PNEUMO
  PNEUMO -> DEAD

  GEST_AGE -> IVH
  GEST_AGE -> DEAD
  GEST_AGE -> BWT
  GEST_AGE -> APGAR5
  GEST_AGE -> PNEUMO

  BWT -> IVH
  BWT -> DEAD

  ACS -> IVH
  ACS -> DEAD

  CHORIO -> GEST_AGE
  CHORIO -> IVH
  CHORIO -> DEAD

  DELIVERY -> IVH
  DELIVERY -> DEAD

  APGAR5 -> IVH
  APGAR5 -> DEAD

  SEX -> IVH
  SEX -> DEAD
}

```

DAGitty analysis (for total effect of IVH on death): - Minimal sufficient adjustment set identified by this DAG:

{GEST_AGE, BWT, ACS, CHORIO, DELIVERY, APGAR5, SEX} - PNEUMO is modeled as a post-exposure mediator (IVH -> PNEUMO -> DEAD), so it should **not** be adjusted for when estimating the total causal effect of IVH.

3 Comparison with HW1 Model

Compare variables identified in the DAG to the covariates included in the adjusted regression model in Homework 1.

1. Covariates in HW1 adjusted model: pneumo, bwt_cat, gest_cat.
2. DAGitty-based target adjustment set (if sample size is not limiting):
GEST_AGE, BWT, ACS, CHORIO, DELIVERY, APGAR5, SEX.

3. Variables missing from HW1 (relative to DAGitty set):

- ACS (antenatal corticosteroids)
- CHORIO (chorioamnionitis)
- DELIVERY (delivery mode)
- APGAR5 (5-min Apgar)
- SEX (infant sex)

4. Role of pneumothorax in causal modeling:

- In this DAG, pneumothorax is primarily post-exposure and on the pathway IVH → PNEUMO → DEAD.
- Therefore, for estimating the **total effect** of IVH on mortality, pneumothorax should not be adjusted for.
- If the research target were a controlled direct effect (not total effect), mediation-oriented methods would be required.

5. Conclusion for HW2 Question 3:

- Yes, additional confounders should have been included if data and sample size allowed.
- I would add back ACS, CHORIO, DELIVERY, APGAR5, and SEX, and avoid routine adjustment for **pneumo** in the primary total-effect model.

4 E-Values Calculation

Calculate the E-values for the exposure of interest (IVH). Do you think the association is likely to be explained away by unmeasured confounders?

Data from Homework 1 (Poisson Model): * Adjusted Risk Ratio (RR_{obs}): 2.059 (from HW1 final Poisson model) * **95% Confidence Interval:** 1.351 - 3.137

Formula:

$$E\text{-value} = RR + \sqrt{RR \times (RR - 1)}$$

Computed with R (EValue package; confirmed by manual formula):

1. **E-value for point estimate** ($RR = 2.059$):

$$\mathbf{E = 3.536}$$

2. **E-value for lower confidence limit** ($LL = 1.351$):

$$E_{LL} = 2.040$$

Interpretation: * To explain away the observed risk ratio of 2.059, an unmeasured confounder would need to have an association with both IVH and death of at least **RR = 3.536**, conditional on measured covariates. * To move the lower CI bound to the null, an unmeasured confounder would still need an association of at least **RR = 2.040** with both exposure and outcome.

Conclusion: * An E-value of 3.536 is relatively large in this clinical context. Few single unmeasured confounders (e.g., chorioamnionitis with effect sizes often around 1.9-2.2) are likely to be strong enough on both links after adjustment for prematurity-related factors. * Therefore, it is **unlikely** that the observed association between IVH and infant mortality is entirely explained away by unmeasured confounding. The evidence supports IVH as an independent risk factor.

5 References

1. Wei JC, Catalano R, Profit J, Gould JB, Lee HC. Impact of antenatal steroids on intraventricular hemorrhage in very-low-birth weight infants. *J Perinatol.* 2016;36(5):352-356. doi: 10.1038/jp.2016.38
2. Roberts D, Brown J, Medley N, Dalziel SR. Antenatal corticosteroids for accelerating fetal lung maturation for women at risk of preterm birth. *Cochrane Database Syst Rev.* 2017;3(3):CD004454. doi: 10.1002/14651858.CD004454.pub3
3. Villamor-Martinez E, et al. Chorioamnionitis is a risk factor for intraventricular hemorrhage in preterm infants: a systematic review and meta-analysis. *Front Physiol.* 2018;9:1253. doi: 10.3389/fphys.2018.01253
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5. Lee HC, Subeh M, Gould JB. Low Apgar score and mortality in extremely preterm neonates born in the United States. *Acta Paediatr.* 2010;99(12):1785-1789. doi: 10.1111/j.1651-2227.2010.01935.x